

11/10/21

R- PROGRAMMING

TOPIC: BASICS

⇒ Introduction to R:

- It is free, open-source programming language
 - Developed by Ross and Robert
 - Supported on Windows, Linux, macOS / CRAN (packages)
 - Both language and Environment.
- Statistical
Data analysis
Data visualization

⇒ Ways to run R:

1) R Console:

Type and Execute.

2) R Script - Runs .R Files.

2) R Studio:

IDE. - Contains Console, Environment, Packages.

⇒ Features:

- It is open source, Interpreted, Case-sensitive, Vectorized.
- Rich Packages, Good Graphics & Cross-platform.

⇒ R Session functions:

getwd() - current directory.

setwd() - Change

ls() - list variables.

rm() - Delete variable

save.image() - Save workspace.

q() - Quit R.

⇒ Functions:

Reusable block of Code:

```
fun <- function(x) {  
  return(x)  
}
```

Code:

Square ← function(x) {

return (x * x)

}

Square(4)

⇒ Operators:

+, -, *, /, ^, %, %/, %/ %

Functions:

Sqrt()

abs()

round()

log()

sum()

mean()



⇒ Variables:

Assignments:

←

=

→

equal

left → right

Rules:

Start with letter or .

No space, case sensitive.

⇒ Data types:

Numeric - 3.14

Integer - 5L

Raw - as.raw(10)

Complex - 2+3i

Type:

class(x)

typeof(x)

⇒ Data Structures:

Vector	1D, same type
List	mixed data
Matrix	2D, same type
Array	Multi
Factor	Categorical
Data Frame	Table, diff. column types.

Revision:

R = Statistical programming language.

Best IDE = R Studio.

Assign = <-

Main func. = getwd(), setwd().

Data types = Numeric, Integer, Character, Raw.

~~Data Structure~~ = Vector, List, Matrix, Array.

Good

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